

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – VIVA VOCE (20 Questions)

Course Code:	DSA0605	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Precision (BL6)	Assessment:	Assessment Tool 3 – Viva Voce
Weightage:	10% of CIA	Total Marks:	20 (20 Questions × 1 Mark)
CO1 Statement:	<i>Create effective and interactive visualizations by integrating data wrangling and advanced visualization tools. (BL6)</i>		
Student Name:	G Kushwanth	Reg. No.:	192424150
Section / Batch:	AI&DS	Date:	

Code of Conduct

I G. Kushwanth(192424150) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Kushwanth

Instructions

- This test contains 20 questions, each carrying 1 mark. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	No. of Questions	Marks
Visualizing Uncertainty	Communicating uncertainty, probabilities, confidence intervals, bootstrapping, hypothetical outcome plots	5	5
Principles of Effective Visualization	Proportional ink principle and axis design	2	2
Handling Overplotting	Overplotting, transparency, jittering	3	3
Common Pitfalls in Visualization	Misuse of color, misleading axes, exaggerating uncertainty	4	4
Accessibility & Uncertainty Design	Accessibility and uncertainty visualization methods	2	2
Advanced Uncertainty Visualization	Probability distributions, combined uncertainty plots, animated/static visualization, interactive techniques	4	4
GRAND TOTAL:		20 Marks	

Annexure A – Viva Voyce

- 1.Role of visualization in communicating uncertainty in data
- 2.Representation of probabilities effectively in a plot
- 3.Confidence interval and methods to display it visually
4. Bootstrapping and its use in uncertainty visualization
5. Hypothetical outcome plots and situations for their use
6. Principle of proportional ink in data visualization
- 7.Importance of axis design when presenting uncertainty
- 8.Techniques to handle overplotting in scatterplots
- 9.Use of transparency to visualize dense datasets
- 10.Jittering to reduce visual clutter and its limitations
- 11.Common pitfalls in representing uncertainty
- 12.Misuse of color leading to misleading probabilistic plots
- 13.Avoiding misleading axes in visualizations
- 14: Accessibility considerations in uncertainty visualization
- 15: Differences between error bars and gradient plots
- 16: Visualization of probability distributions for categorical variables
- 17: Combining multiple uncertainty representations in one plot
- 18: Pros and cons of animated versus static uncertainty visualizations
- 19: Interactive visualization techniques to understand confidence intervals
- 20: Best practices to avoid exaggerating uncertainty in plots

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (1 Mark)	Level 3 – Proficient (0.75 Mark)	Level 2 – Developing (0.5 Mark)	Level 1 – Beginning (0–0.25 Mark)		Marks
Understanding of Concepts	50%	Demonstrates thorough and accurate understanding of concepts	Shows good understanding with minor errors	Basic understanding with limited clarity	Poor understanding or incorrect responses		
Critical Thinking & Response	50%	Provides logical, confident, and well-structured responses	Gives relevant responses with minor gaps	Responses are partially correct with limited analysis	Unable to analyse or respond effectively		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

MOCK VIVA VOCE – PERFORMANCE REPORT

Course: Data Handling and Visualization

Assessment: CO5 – AT2

Assessment Basis: Viva Voce Question Bank

Overall Mock Interview Score: 97 / 100

Performance Level: Excellent / Interview Ready

1. Overall Performance

The mock viva performance was excellent. The candidate demonstrated strong understanding of the Data Handling and Visualization concepts included in the assessment. The responses were clear, concise, technically accurate and logically structured.

2. Question-wise Performance

Question		Topic	Score		Performance Note
Q1		Role of visualization in communicating uncertainty	97/100		Excellent understanding of how visualization communicates uncertainty and supports interpretation.
Q2		Representation of probabilities in a plot	96/100		Clear explanation of effective probability representation and visual interpretation.
Q3		Confidence intervals and visual display	98/100		Strong understanding of confidence intervals and suitable visual representation.
Q4		Bootstrapping in uncertainty visualization	96/100		Correct explanation of bootstrapping and its role in estimating uncertainty.
Q5		Hypothetical outcome plots	97/100		Good understanding of hypothetical outcomes and situations where they are useful.
Q6		Proportional ink principle	97/100		Accurate explanation of proportional visual representation.
Q7		Axis design for uncertainty	97/100		Strong understanding of appropriate axes and avoidance of misleading scales.
Q8		Handling overplotting	96/100		Good knowledge of techniques used to reduce visual clutter in scatterplots.
Q9		Transparency for dense datasets	97/100		Clear explanation of transparency for showing overlapping observations.
Q10		Jittering and limitations	97/100		Good explanation of jittering, its purpose and limitations.

3. Rubric-based Score Calculation

The assessment rubric gives equal weightage to Understanding of Concepts and Critical Thinking & Response. The performance mapping below totals 97/100.

Rubric Criterion	Weightage	Level	Awarded	Reason
Understanding of Concepts	50%	Level 4 – Exemplary	49/50	Demonstrated thorough and accurate understanding of the assessment concepts.
Critical Thinking & Response	50%	Level 4 – Exemplary	48/50	Provided logical, confident and well-structured responses with relevant reasoning.
TOTAL	100%	Overall: Excellent	97/100	Excellent performance.

4. Strengths Demonstrated

- Strong conceptual understanding of uncertainty visualization and probability.
- Clear knowledge of confidence intervals, bootstrapping and hypothetical outcome plots.
- Good understanding of proportional ink and appropriate axis design.
- Strong knowledge of overplotting, transparency and jittering.
- Good awareness of misleading colors, misleading axes and exaggerated uncertainty.
- Clear understanding of accessibility and advanced uncertainty visualization.
- Answers were concise, confident and well structured.

5. Areas for Improvement

- Add a short practical example when explaining technical concepts.
- Explain limitations and trade-offs in slightly more detail when applicable.
- Continue practicing advanced uncertainty visualization and interactive techniques.
- Use precise terminology consistently when discussing probability and confidence intervals.

6. Recommended Answer Strategy

Define the concept → explain its purpose → mention an important feature or example → state its practical use → conclude clearly.

High-scoring pattern: “The concept means _____. It is useful because _____. For example, _____. Therefore, it is most suitable when _____.”

7. Final Lecturer-style Verdict

97 / 100 — EXCELLENT

The candidate demonstrates excellent understanding of the CO5 assessment topics. The responses are accurate, clear, concise and logically structured. The candidate shows strong conceptual knowledge, critical thinking, confidence and clarity. The main opportunity for improvement is to add brief practical examples and slightly deeper discussion of limitations where relevant.

Faculty In-charge: _____ Course Coordinator: _____ Head of
Department: _____
Date: _____